Coordinate-power middle fibers contain a full analytic ball

A solution of the highlighted example in arXiv:2102.06771

Candidate solution packet

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Status. Candidate full resolution of the highlighted coordinate-square example, likely valid pending expert review.

1 Source question

Let B_{ℓ_2} be the open unit ball of the complex Hilbert space ℓ_2. Dimant and Singer study the vector-valued spectrum

$$\mathcal{M}_{u,\infty}(B_{\ell_2}, B_{\ell_2}) = \{\Phi : \mathcal{A}_u(B_{\ell_2}) \longrightarrow \mathcal{H}^\infty(B_{\ell_2}) : \Phi \text{ is a nonzero algebra homomorphism}\}.$$

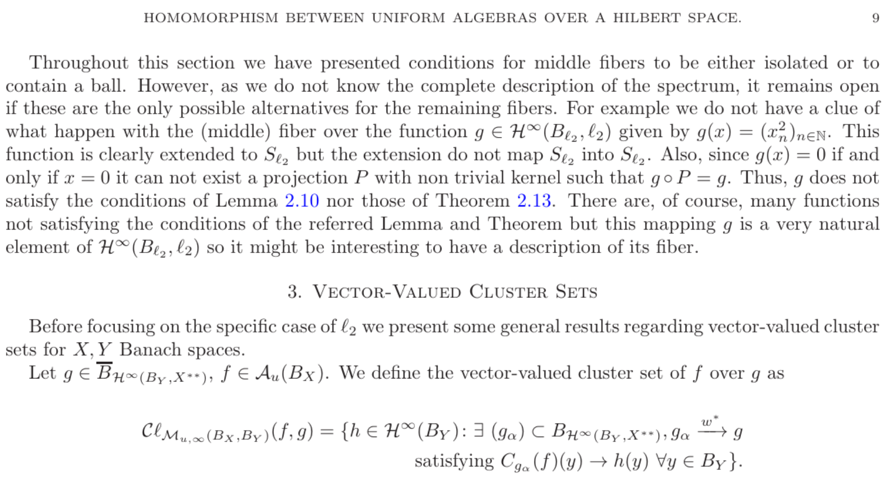
It has the natural projection

$$\xi(\Phi)(x) = \big(\Phi(\langle\,\cdot, e_n\rangle)(x)\big)_{n\geq 1} \quad \text{and fibers} \quad \mathscr{F}(g) = \xi^{-1}(g).$$

Their Section 2 shows that some middle fibers are singletons while others contain an analytic copy of a ball. At the end of the section they ask what happens for the natural middle symbol

$$g_2(x) = (x_n^2)_{n\geq 1},$$

which satisfies neither of their sufficient criteria [1].



HOMOMORPHISM BETWEEN UNIFORM ALGEBRAS OVER A HILBERT SPACE. 9

Throughout this section we have presented conditions for middle fibers to be either isolated or to contain a ball. However, as we do not know the complete description of the spectrum, it remains open if these are the only possible alternatives for the remaining fibers. For example we do not have a clue of what happen with the (middle) fiber over the function $g \in \mathcal{H}^\infty(B_{\ell_2}, \ell_2)$ given by $g(x) = (x_n^2)_{n\in\mathbb{N}}$. This function is clearly extended to S_{ℓ_2} but the extension do not map S_{ℓ_2} into S_{ℓ_2}. Also, since $g(x) = 0$ if and only if $x = 0$ it can not exist a projection P with non trivial kernel such that $g \circ P = g$. Thus, g does not satisfy the conditions of Lemma 2.10 nor those of Theorem 2.13. There are, of course, many functions not satisfying the conditions of the referred Lemma and Theorem but this mapping g is a very natural element of $\mathcal{H}^\infty(B_{\ell_2}, \ell_2)$ so it might be interesting to have a description of its fiber.

3. Vector-Valued Cluster Sets

Before focusing on the specific case of ℓ_2 we present some general results regarding vector-valued cluster sets for X, Y Banach spaces.

Let $g \in \overline{B}_{\mathcal{H}^\infty(B_Y, X^{**})}$, $f \in \mathcal{A}_u(B_X)$. We define the vector-valued cluster set of f over g as

$$\mathcal{C}\ell_{\mathcal{M}_{u,\infty}(B_X,B_Y)}(f,g) = \{h \in \mathcal{H}^\infty(B_Y) \colon \exists\, (g_\alpha) \subset B_{\mathcal{H}^\infty(B_Y,X^{**})}, g_\alpha \xrightarrow{w^*} g \text{ satisfying } C_{g_\alpha}(f)(y) \to h(y)\ \forall y \in B_Y\}.$$

Figure 1: The highlighted unresolved example at the end of Section 2, source PDF page 9.

The source passage asks for a description of this fiber and places it in the broader question whether singleton fibers and fibers containing analytic balls are the only possible middle-fiber behaviors.

2 Proof intuition

The obstruction to applying the source's ball criterion is that g_2 loses no input direction. Instead of looking for a kernel direction, one can find norm room intrinsic to the coordinate-power map. The fixed holomorphic factor $\phi_m(x) = x_1 \cdots x_m$ satisfies

$$\|g_m(x)\|^2 + |\phi_m(x)|^2 \leq \|x\|^{2m}, \qquad g_m(x) = (x_n^m)_n.$$

Thus an arbitrary bounded holomorphic vector map can be multiplied by ϕ_m and placed in coordinates escaping to infinity without leaving the Hilbert ball. Weak-star ultrafilter limits of the corresponding composition homomorphisms have projection g_m.

The escaping coordinates might appear to erase the parameter. They do not: diagonal quadratic and cubic polynomials convert those tails into generating functions for the coordinate squares and cubes of the parameter. Their boundary values determine all coefficients. Squares and cubes together remove the sign ambiguity and recover the parameter exactly.

3 Main result

For $k \geq 1$, let P_k be the orthogonal projection onto $\operatorname{span}\{e_1, \dots, e_k\}$ and let $S_k e_j = e_{k+j}$.

Theorem 1. *For every integer $m \geq 2$, put*

$$g_m(x) = (x_n^m)_{n\geq 1} \qquad (x \in B_{\ell_2}).$$

Then $\|g_m\| = 1$, $g_m(B_{\ell_2}) \subset B_{\ell_2}$, and there is an analytic injection

$$\Psi_m : B_{\mathcal{H}^\infty(B_{\ell_2},\ell_2)} \longrightarrow \mathscr{F}(g_m).$$

Moreover $\Psi_m(0) = C_{g_m}$. In particular, the source's highlighted fiber over $(x_n^2)_n$ contains an analytic copy of the full open unit ball of $\mathcal{H}^\infty(B_{\ell_2}, \ell_2)$.

Proof. Fix $m \geq 2$ and define

$$\phi_m(x) = x_1 x_2 \cdots x_m.$$

If $a_n = |x_n|^2$, the multinomial expansion (first for finite partial sums, then by monotone convergence) gives

$$\sum_{n=1}^{\infty} a_n^m + a_1 a_2 \cdots a_m \leq \left(\sum_{n=1}^{\infty} a_n\right)^m. \tag{1}$$

Indeed, the terms a_n^m occur with coefficient one and the mixed monomial $a_1 \cdots a_m$ occurs with coefficient $m!$. Consequently

$$\|g_m(x)\|^2 + |\phi_m(x)|^2 \leq \|x\|^{2m} < 1 \qquad (x \in B_{\ell_2}). \tag{2}$$

This also shows that $g_m(B_{\ell_2}) \subset B_{\ell_2}$; testing points re_1 as $r \uparrow 1$ shows that $\|g_m\| = 1$.

For $h \in B_{\mathcal{H}^\infty(B_{\ell_2},\ell_2)}$ and $k \geq 1$, define

$$s_{h,k}(x) = P_k g_m(x) + \phi_m(x) S_k h(x).$$

The two summands have orthogonal coordinate supports. Hence, by (2),

$$\|s_{h,k}(x)\|^2 = \|P_k g_m(x)\|^2 + |\phi_m(x)|^2 \|h(x)\|^2 \leq \|g_m(x)\|^2 + |\phi_m(x)|^2 < 1.$$

Thus $s_{h,k} : B_{\ell_2} \to B_{\ell_2}$ is holomorphic and defines the composition homomorphism

$$\Psi_{m,k}(h) = C_{s_{h,k}} \in \mathcal{M}_{u,\infty}(B_{\ell_2}, B_{\ell_2}).$$

Fix a free ultrafilter $\mathscr{U}$ on $\mathbb{N}$ containing every cofinite set, and set

$$\Psi_m(h) = \lim_{k\to\mathscr{U}} \Psi_{m,k}(h)$$

in the weak-star compact vector-valued spectrum. The limit is an algebra homomorphism. The standard bounded-holomorphic ultrafilter argument used in [1, Lemma 2.1, Theorem 2.3, Example 2.12] shows that Ψ_m is analytic: for each $f \in \mathcal{A}_u(B_{\ell_2})$ and $x \in B_{\ell_2}$, the scalar maps $h \mapsto f(s_{h,k}(x))$ are bounded holomorphic functions, and their weak-star ultrafilter limit is holomorphic.

For every fixed n and every $k \geq n$,

$$\Psi_{m,k}(h)(\langle\,\cdot, e_n\rangle)(x) = x_n^m.$$

Therefore $\xi(\Psi_m(h)) = g_m$, so the image of Ψ_m lies in $\mathscr{F}(g_m)$.

It remains to prove injectivity. For $q \in \{2, 3\}$ and $\lambda \in \mathbb{T}$, consider the continuous homogeneous polynomial

$$F_{q,\lambda}(y) = \sum_{j=1}^{\infty} \lambda^j y_j^q \qquad (y \in \ell_2).$$

It belongs to $\mathcal{A}_u(B_{\ell_2})$. Put $\chi(\lambda) = \lim_{k\to\mathscr{U}} \lambda^k$; since $|\lambda| = 1$, $|\chi(\lambda)| = 1$. Direct substitution and passage to the ultrafilter limit give, for $x \in B_{\ell_2}$,

$$\Psi_m(h)(F_{q,\lambda})(x) = \sum_{n=1}^{\infty} \lambda^n x_n^{mq} + \chi(\lambda)\phi_m(x)^q \sum_{j=1}^{\infty} \lambda^j h_j(x)^q. \tag{3}$$

Suppose $\Psi_m(h) = \Psi_m(h')$. On the nonempty open set

$$V_m = \{x \in B_{\ell_2} : \phi_m(x) \neq 0\},$$

equation (3), first with $q = 2$ and then with $q = 3$, implies for every $\lambda \in \mathbb{T}$ that

$$\sum_{j\geq 1} \lambda^j h_j(x)^q = \sum_{j\geq 1} \lambda^j h'_j(x)^q.$$

These power series have absolutely summable coefficients and are continuous on the closed disk. Uniqueness of their Taylor coefficients yields

$$h_j(x)^2 = h'_j(x)^2, \qquad h_j(x)^3 = h'_j(x)^3 \quad (j \geq 1).$$

For complex numbers, equality of both the square and cube forces equality; hence $h_j(x) = h'_j(x)$ for every j and every $x \in V_m$. Each coordinate of $h - h'$ is holomorphic, so the identity theorem on the connected ball B_{ℓ_2} gives $h = h'$ everywhere. Thus Ψ_m is injective.

Finally, when $h = 0$ one has $s_{0,k} = P_k g_m$. For every $x \in B_{\ell_2}$, $P_k g_m(x) \to g_m(x)$ in norm. Since every $f \in \mathcal{A}_u(B_{\ell_2})$ is uniformly continuous, $f(P_k g_m(x)) \to f(g_m(x))$; therefore $\Psi_m(0) = C_{g_m}$. □

4 Consequences and limitations

Taking $m = 2$ fully determines which of the two known large-scale behaviors occurs for the source paper's singled-out example: its fiber is not a singleton, but contains the same full analytic parameter ball appearing in the source's large-fiber results. The theorem simultaneously treats every coordinate power $m \geq 2$.

The broader classification question remains open. This argument does not show that every middle fiber is either a singleton or contains an analytic ball, and therefore does not exclude a third behavior for other symbols.

5 Verification and novelty status

The proof was audited at the following points: the multinomial norm estimate; strict inclusion $s_{h,k}(B_{\ell_2}) \subset B_{\ell_2}$; coordinatewise fiber projection; membership of the quadratic and cubic tests in $\mathcal{A}_u(B_{\ell_2})$; nonvanishing of the ultrafilter phase on $\mathbb{T}$; coefficient recovery; and extension from V_m by the identity theorem. The only imported mechanism is the source paper's own weak-star ultrafilter construction of analytic maps into the spectrum.

On August 11, 2026, the local run indexes were searched by arXiv id, exact title, the displayed coordinate-square example, and middle-fiber terminology. Targeted web searches found the source arXiv record and its 2022 journal publication, but no later answer to this example. The search was bounded, so novelty confidence is moderate pending a specialist citation review.

References

[1] V. Dimant and J. Singer, *A look into homomorphisms between uniform algebras over a Hilbert space*, Studia Math. **265** (2022), no. 1, 57–75; arXiv:2102.06771; DOI: 10.4064/sm210219-10-6.